A09 POLLUTION INFILTRATION MANAGEMENT | O (MAX : 2 PT)

Intent : Minimize the introduction of pollutants into indoor air through the building envelope and at building entrances.

Summary : This WELL feature requires projects to reduce transmission of air and pollutants from outdoors to indoors through the building envelope and entrance.

Issue : Research shows that approximately 65% of outdoor air particle inhalation occurs while indoors.[1] Exposure to high levels of coarse and fine particulate matter inadvertently introduced into the space can lead to respiratory irritation. It has been associated with increases in lung cancer, cardiovascular disease and mortality.[2] Indoor air quality and thermal comfort can be compromised by leaks and gaps that break the building's air barrier. These weak points are not only wasteful from an energy point of view but can also lead to conditions conducive to mold growth and to the infiltration of pests or polluted air. In addition, building users can introduce particulate matter indoors through their clothes and shoes, including harmful coliforms and *Escherichia coli*, among other toxins.[3–5]

Solutions : In addition to building envelope commissioning, there is a need for measures that minimize or prevent the introduction of potentially harmful substances into indoor spaces. An example of such interventions is the installation of entryway walk-off systems and/or entryway air seals at all main building entrances.[6]

PART 1 DESIGN HEALTHY ENTRYWAYS (MAX : 1 PT)

For All Spaces:

1: Building entry design

For any functional building entrance (not including doors to balconies or terraces), the following design features are present:

a. At least one of the following strategies are in place at entryways that span, at minimum, the width of the entrance and 3 m long in the primary direction of travel (sum of indoor and outdoor length):[7]
   1. Grilles.
   2. Grates or slots.
   3. Rollout mats.
   4. Removable carpet tiles.
   5. Any other product designed to remove dirt from shoes at the entrance.

b. At least one of the following strategies are in place to slow the movement of air from outdoors to indoors:
   1. Building entry vestibule with two typically closed doorways.
   2. Revolving entrance doors.
   3. Buildings with an entrance that is outside of the project boundary, or buildings with an entrance lobby that is not regularly occupied, must have at least three typically shut doors that separate the outdoors from all regularly occupied spaces within the project boundary.
   4. Air curtains installed and commissioned in accordance with ASHRAE Standard 90.1-2019.

2: Building entry maintenance

Building entryway systems are cleaned, as follows:

a. Wet-cleaned at least once a week, or as instructed by manufacturer.
b. Vacuumed at least once a day, or as instructed by manufacturer.

For Outdoor Sport Areas:

The following requirement is met:

a. All facilities adjacent to an outdoor sports field have an area (e.g., staging area, mudroom, drying room) that separates the playing field from other internal areas to capture moisture and debris.

Note :

Interiors projects may count exterior building entrances that meet the feature requirements, even if they are outside of the project boundary.

PART 2 PERFORM ENVELOPE COMMISSIONING (MAX : 1 PT)

For All Spaces:

For projects undergoing design and construction, the following requirements are met:

a. The project uses a façade engineer that is responsible for defining the building envelope performance metrics (including materials, components, assemblies and systems) at the concept design stage.
b. The building envelope performance requirements are included in the Basis of Design document and reflect the Owner's Project Requirements.
c. The commissioning process includes envelope commissioning for air infiltration and leakage, which is reflected in the specification and commissioning plan.
d. The envelope commissioning process is executed, as outlined in the commissioning plan.
e. The envelope commissioning plan is included in the project Operation & Maintenance (O&M) Manual.

Note :

Interiors projects may count commissioning performed on the building envelope, even if outside of the project boundary.

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